

OPERATING SYSTEMS (CS304)

Process States & Transition

Learning Objectives

Understand the concept of processes, process states, state transitions, Process Control Block (PCB), and process scheduling.

Introduction to Processes

A process is a program in execution. It consists of program code, data, stack, heap, and execution context maintained by the operating system.

What is a Process?

A process is an active entity that requires CPU time, memory, and other system resources for execution.

Characteristics of a Process

- Dynamic in nature
- Has its own address space
- Requires system resources
- Can interact with other processes

Process States

1. New
2. Ready
3. Running
4. Waiting (Blocked)
5. Terminated

New State

The process is being created and initialized by the operating system.

Ready State

The process is waiting in the ready queue to be assigned CPU time.

Running State

The process is currently executing instructions on the CPU.

Waiting (Blocked) State

The process is waiting for an event such as I/O completion or resource availability.

Terminated State

The process has completed execution or has been terminated by the operating system.

Process State Transitions

New → Ready
Ready → Running
Running → Waiting
Waiting → Ready
Running → Terminated

Process Control Block (PCB)

The PCB is a data structure used by the operating system to store information about a process such as process ID, state, CPU registers, memory information, and scheduling details.

Importance of PCB

PCB enables process management, context switching, scheduling, and resource allocation.

Context Switching

Context switching is the process of saving the state of one process and loading the state of another process for CPU execution.

Advantages of Process Management

- Efficient CPU utilization
- Better resource allocation
- Supports multitasking
- Improves system responsiveness

Exam Important Questions

1. Define Process.
2. Explain different Process States.
3. Draw and explain the Process State Transition Diagram.
4. What is PCB?
5. Explain Context Switching.

Summary

Processes are fundamental components of an operating system. Understanding process states, transitions, PCB, and context switching is essential for learning process management.